produces heat. And immediately we are able to distinguish the future (towards which the pendulum slows) from the past. We have never seen a pendulum start swinging from a stationary position, with its movement initiated by the energy obtained by absorbing heat from its supports. The difference between past and future only exists when there is heat. The fundamental phenomenon that distinguishes the future from the past is the fact that heat passes from things that are hotter to things that are colder.

So, again, why, as time goes by, does heat pass from hot things to cold and not the other way round?

The reason was discovered by Boltzmann, and is surprisingly simple: *it is sheer chance*.

Boltzmann's idea is subtle, and brings into play the idea of probability. Heat does not move from hot things to cold things due to an absolute law: it only does so with a large degree of probability. The reason for this is that it is statistically more probable that a quickly moving atom of the hot substance collides with a cold one and leaves it a little of its energy, rather than vice versa. Energy is conserved in the collisions, but tends to get distributed in more or less equal parts when there are many collisions. In this way the temperature of objects in contact with each other tends to equalize. It is not impossible for a hot body to become hotter through contact with a colder one: it is just extremely improbable.

This bringing of *probability* to the heart of physics, and using it to explain the bases of the dynamics of heat, was initially considered to be absurd. As frequently happens, no one took Boltzmann seriously. On 5 September 1906, in Duino near Trieste, he committed suicide by hanging himself, never having witnessed the subsequent universal recognition of the validity of his ideas.

In the second lesson I related how quantum mechanics predicts that the movement of every minute thing occurs by chance. This puts probability into play as well. But the probability which Boltzmann considered, the probability at the roots of heat, has a different nature, and is independent of quantum mechanics. The probability in play in the science of heat is in a certain sense tied to our *ignorance*.

I may not know something with certainty, but I can still assign a lesser or greater degree of probability to something. For instance, I don't know whether it will rain tomorrow here in Marseilles, or whether it will be sunny or will snow, but the probability that it will snow here tomorrow – in Marseilles, in August – is low. Similarly with regard to most physical objects: